Finiteness and Improved Uniform Bounds for Signed Powers of Integer Linear Polynomials

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Abstract

Let $F(k,n)$ be the least number of terms in a polynomial identity

$$n=\sum_{i=1}^{s}\epsilon_i(a_ix+b_i)^k,\qquad \epsilon_i\in\{-1,1\},\quad a_i,b_i\in\mathbb{Z},\quad a_i\neq 0.$$

We prove that $F(k,n)$ is finite for every integer $k>1$ and every $n\in\mathbb{Z}$, giving an elementary factorial bound by finite differences. We then compress the binomial coefficients in the linearization identity using the easier Waring problem and obtain the sharper general estimate $F(k,n)\le v(k)C_k\le(k+1)v(k)^2$. Published upper bounds for the easier Waring number yield explicit bounds for $4\le k\le 8$, while classical Waring estimates imply $F(k,n)=O(k^3(\log k)^2)$ uniformly in n. Finally, we prove the sharp quadratic result $\sup_{n\in\mathbb{Z}}F(2,n)=4$. The problem and earlier specialized bounds for $k=2,3$ were discussed in a MathOverflow post; that post is explicitly cited below. To avoid duplication, its nine-term quadratic argument and its specialized 37-term cubic argument are not reproduced or claimed here.

Keywords: easier Waring problem; finite differences; polynomial identities; signed sums of powers.
MSC 2020: 11P05, 11D99.

1 Problem, notation, and provenance

The motivating question is whether every integer can be represented as a signed sum of powers of nonconstant integer linear polynomials. The question, the notation $F(k,n)$, and earlier bounds for the cases $k=2,3$ appear in Tong's MathOverflow post [1]. The present note is written with that prior discussion in view.

For integers $k>1$ and n, define

$$F(k,n)=\min\left\{s\ge 1: n=\sum_{i=1}^{s}\epsilon_i(a_ix+b_i)^k \text{ in } \mathbb{Z}[x]\right\},$$

where $\epsilon_i\in\{-1,1\}$ and $a_i,b_i\in\mathbb{Z}$ with $a_i\neq 0$. Repetitions are allowed, and equality means equality as polynomials, not equality only at chosen values of x.

For this note, let $v(k)$ denote the least number such that every integer is a signed sum of at most $v(k)$ integer k-th powers with nonzero bases. Let $r_k(c)$ be the least number of such signed powers needed to represent a specified integer c; set $r_k(0)=0$. These are the usual easier-Waring quantities, in the convention of allowing signed terms; see Wright [5] and Revoy [2].

Overlap and attribution. Tong's post already contains the claim $F(2,n) \le 9$, the specialized claim $F(3,n) \le 37$, and the Choudhry/Rao identities and residue-class construction used for the latter. Those proofs and calculations are deliberately omitted here. The coarse general bound in Theorem 3.3 was also stated in a comment on the same MathOverflow discussion; the purpose here is to give a complete proof and place it in a sharper framework. The coefficient-compression argument, the resulting numerical table, and the proof of the sharp quadratic bound are developed in the present note. No claim of priority over unsearched or unpublished work is intended.

2 Finite-difference linearization

Lemma 2.1 (Linearization identity)**.** *For every integer $k \ge 1$ and every $m \in \mathbb{Z}$,*

$$m^k = \sum_{j=1}^{k+1} (-1)^{j-1} \binom{k+1}{j} (jx+m)^k. \tag{2.1}$$

Proof. Expand the right-hand side by the binomial theorem. For $0 \le r \le k$, the coefficient of x^r is

$$\binom{k}{r} m^{k-r} \sum_{j=1}^{k+1} (-1)^{j-1} \binom{k+1}{j} j^r.$$

When $r = 0$, the inner sum is 1, since $\sum_{j=1}^{k+1} (-1)^{j-1} \binom{k+1}{j} = 1$. For $1 \le r \le k$, the inner sum is zero: it is the negative of the full $(k+1)$-st finite-difference sum for the polynomial t^r, and the $j = 0$ term is zero. Thus the constant coefficient is m^k and every coefficient of positive degree vanishes. This proves (2.1). □

Corollary 2.2 (Uncompressed linearization cost)**.** *Every constant m^k is a signed sum of at most*

$$L_k = \sum_{j=1}^{k+1} \binom{k+1}{j} = 2^{k+1} - 1$$

powers of integer linear polynomials with nonzero slopes.

Proof. Replace each coefficient $\binom{k+1}{j}$ in (2.1) by that many copies of its term, with the corresponding sign. Each polynomial $jx + m$ has slope $j \ne 0$, and the total number of terms is L_k. □

3 An elementary uniform bound

Lemma 3.1 (A finite-difference progression)**.** *For $k \ge 2$, put*

$$S_k(t) = \sum_{j=0}^{k-1} (-1)^{k-1-j} \binom{k-1}{j} (t+j)^k.$$

There is an integer A_k such that $S_k(t) = k!t + A_k$ for every integer t.

Proof. The expression is the $(k-1)$-st forward difference of t^k. Each forward difference lowers the degree by one; after $k-1$ differences the polynomial is linear, with leading coefficient successively multiplied by $k, k-1, \dots, 2$. The resulting leading coefficient is $k!$. The constant term $A_k = S_k(0)$ is an integer. □

Theorem 3.2 (Elementary easier-Waring bound)**.** *Every integer is a signed sum of at most* $2^k+k!-1$ *integer* k*-th powers. In particular,*

$$\boxed{v(k) \le 2^k + k! - 1.}$$

Proof. Lemma 3.1 gives $k!q = S_k(q) - S_k(0)$. Each of the two expressions $S_k(q)$ and $S_k(0)$ uses at most $\sum_{j=0}^{k-1} \binom{k-1}{j} = 2^{k-1}$ signed powers when each binomial coefficient is implemented by repeated terms. Their difference therefore uses at most 2^k terms. Write $n = k!q + r$ with $0 \le r \le k! - 1$ and add r copies of 1^k. Any zero bases can be discarded. For $n = 0$, use $0 = 1^k - 1^k$ if a nonempty representation is desired. □

Theorem 3.3 (Coarse uniform bound)**.** *For all* $k > 1$ *and all* $n \in \mathbb{Z}$*,*

$$\boxed{F(k,n) \le (2^{k+1}-1)(2^k + k! - 1).} \tag{3.1}$$

Proof. By Theorem 3.2, write n as a signed sum of at most $2^k + k! - 1$ integer k-th powers. Apply Lemma 2.1 to each power and use Corollary 2.2. Every resulting linear polynomial has nonzero slope, and multiplying the two term-count bounds gives (3.1). □

The first values of the bound are as follows. The last column is obtained by multiplying the middle two columns.

k	$2^{k+1}-1$	$2^k + k! - 1$	product
2	7	5	35
3	15	13	195
4	31	39	1209
5	63	151	9513

In particular, $31 \cdot 39 = 1209$ for $k = 4$.

4 Coefficient compression

The coefficients in the linearization identity need not be implemented by simply repeating a term. They may themselves be written as short signed sums of k-th powers.

Theorem 4.1 (Coefficient-compressed bound)**.** *Define*

$$C_k = \sum_{j=1}^{k+1} r_k\left(\binom{k+1}{j}\right). \tag{4.1}$$

Then for every $k > 1$ *and* $n \in \mathbb{Z}$*,*

$$\boxed{F(k,n) \le v(k) C_k \le (k+1) v(k)^2.} \tag{4.2}$$

Proof. Write $n = \sum_{i=1}^{t} \delta_i m_i^k$ with $t \le v(k)$. For each j, choose a representation

$$\binom{k+1}{j} = \sum_{\ell=1}^{r_k\left(\binom{k+1}{j}\right)} \eta_{j,\ell} u_{j,\ell}^k, \qquad u_{j,\ell} \ne 0.$$

Multiplying by $(jx + m_i)^k$ gives

$$\binom{k+1}{j}(jx+m_i)^k = \sum_{\ell} \eta_{j,\ell}\big(u_{j,\ell}(jx+m_i)\big)^k.$$

Every resulting slope is $u_{j,\ell} j \neq 0$. Substitution into (2.1) shows that each m_i^k costs at most C_k admissible terms, proving $F(k,n) \leq v(k)C_k$. There are $k+1$ summands in (4.1), and each is at most $v(k)$, so $C_k \leq (k+1)v(k)$. This proves (4.2). □

Lemma 4.2 (A coefficient-specific bound)**.** *If $1 \leq c \leq 2^k$, then*

$$\boxed{r_k(c) \leq \min\{c,\, 2^k - c + 1,\, v(k)\}.} \tag{4.3}$$

Proof. The representation of c as c copies of 1^k uses c terms. Also,

$$c = 2^k - (2^k - c) \cdot 1^k,$$

which uses one copy of $2^k = (2)^k$ and $2^k - c$ negative copies of 1^k, for a total of $2^k - c + 1$ terms. Finally, the definition of $v(k)$ gives $r_k(c) \leq v(k)$. Taking the minimum proves the claim. □

For every $1 \leq j \leq k$, the binomial coefficient $\binom{k+1}{j}$ is at most 2^k (the largest entry in row $k+1$ of Pascal's triangle is at most half the row sum). Thus Lemma 4.2 applies to all terms of C_k.

4.1 Explicit numerical bounds

Revoy's survey of the easier Waring problem gives, among other bounds, $v(2) = 3$, $v(3) \leq 5$, $v(4) \leq 10$, $v(5) \leq 10$, $v(6) \leq 14$, and $v(8) \leq 28$ [2]. For $k = 7$, Wooley proved $G(7) \leq 31$ for the classical Waring number [4]; Proposition 4.3 below then yields $v(7) \leq 62$. Using Lemma 4.2 term by term gives the following uniform bounds. They are upper bounds, not claims of optimality.

k	Upper bound for $v(k)$	Bound for C_k	Bound for $F(k,n)$
2	3	5	15
3	5	12	60
4	10	25	250
5	10	43	430
6	14	71	994
7	62	244	15128
8	28	187	5236

For example, for $k = 5$ the binomial coefficients are $6, 15, 20, 15, 6, 1$. Formula (4.3) bounds their costs by $6, 10, 10, 10, 6, 1$, whose sum is 43. The other rows are obtained in the same way. In the cubic case, the specialized bound $F(3,n) \leq 37$ reported in [1] is stronger than the generic estimate $F(3,n) \leq 60$ here; its proof is not duplicated in this note.

Proposition 4.3. *Let $G(k)$ be the least integer such that every sufficiently large positive integer is a sum of at most $G(k)$ positive integer k-th powers. Then*

$$v(k) \leq 2G(k).$$

Proof. Fix $n \in \mathbb{Z}$. Choose N large enough that both N and $N+n$ lie in the range covered by the definition of $G(k)$. Represent each as a sum of at most $G(k)$ positive k-th powers and subtract the representation of N from that of $N+n$. The difference is a signed sum of at most $2G(k)$ powers and equals n. □

4.2 Asymptotic consequence

The classical Waring problem has the estimate $G(k) = O(k \log k)$ as $k \to \infty$; see, for example, the survey by Vaughan and Wooley [3] and subsequent work cited there. Hence Proposition 4.3 and Theorem 4.1 imply

$$F(k, n) \le (k+1)v(k)^2 \le 4(k+1)G(k)^2.$$

Consequently, uniformly in n,

$$\boxed{F(k, n) = O\Big(k^3 (\log k)^2\Big).} \tag{4.4}$$

The implied constant is independent of n. This asymptotic estimate uses deep information about the classical Waring problem; the elementary factorial bound in Theorem 3.3 does not.

5 The sharp quadratic result

We now prove

$$\boxed{\sup_{n \in \mathbb{Z}} F(2, n) = 4.}$$

The question of whether a four-term bound holds was raised in the MathOverflow discussion [1]; the proof below addresses that question directly.

Theorem 5.1. *For every $n \in \mathbb{Z}$, one has $F(2, n) \le 4$.*

Proof. Suppose first that n is odd. Since $(n+1)/2$ and $(n-1)/2$ are integers,

$$\begin{aligned} n = (x+1)^2 + \left(2x - \frac{n+1}{2}\right)^2 \\ - (x-1)^2 - \left(2x - \frac{n-1}{2}\right)^2. \end{aligned}$$

Indeed, $(x+1)^2 - (x-1)^2 = 4x$. Setting $A = 2x - (n+1)/2$, the other difference is $A^2 - (A+1)^2 = -2A - 1 = n - 4x$. Their sum equals n.

If n is even, then $n/2 \in \mathbb{Z}$ and

$$\begin{aligned} n = (x+2)^2 + \left(3x + 1 - \frac{n}{2}\right)^2 \\ - (x-1)^2 - \left(3x + 2 - \frac{n}{2}\right)^2. \end{aligned}$$

Here $(x+2)^2 - (x-1)^2 = 6x + 3$. Setting $B = 3x + 1 - n/2$, the other difference is $B^2 - (B+1)^2 = -2B - 1 = n - 6x - 3$. Again the sum equals n. Every displayed polynomial has integer coefficients and nonzero slope. Thus $F(2, n) \le 4$. □

Theorem 5.2 (Sharpness)**.** *The uniform bound in Theorem 5.1 is sharp: $F(2, 2) = 4$, and therefore* $\sup_{n \in \mathbb{Z}} F(2, n) = 4$.

Proof. A single signed square of a nonconstant linear polynomial cannot be constant, because its x^2 coefficient is nonzero. A sum of two squares cannot have vanishing x^2 coefficient. For a difference of two squares, write

$$(ax + u)^2 - (bx + v)^2 = n, \qquad ab \neq 0.$$

Comparing x^2 coefficients gives $a^2 = b^2$. Comparing linear coefficients gives $au = bv$. If $b = a$, then $u = v$; if $b = -a$, then $u = -v$. In either case the constant is $n = 0$. Hence $n = 2$ has no representation with at most two terms.

For three terms, after possibly reversing all signs, the only remaining sign pattern is

$$(ax + u)^2 + (bx + v)^2 - (cx + w)^2 = n, \qquad abc \neq 0.$$

Comparing coefficients yields $a^2 + b^2 = c^2$ and $au + bv = cw$. It follows that

$$\begin{aligned} c^2 n &= c^2(u^2 + v^2) - c^2 w^2 \\ &= (a^2 + b^2)(u^2 + v^2) - (au + bv)^2 \\ &= (av - bu)^2. \end{aligned}$$

Thus n is a nonnegative rational square, hence an integer square. Reversing all signs shows that the other three-term sign pattern can yield only the negative of a square. Since 2 is neither a square nor the negative of a square, $F(2, 2) \geq 4$. Theorem 5.1 gives $F(2, 2) \leq 4$, proving equality and sharpness. □

6 Conclusion and scope

The results proved in this note are:

1. The finite-difference identity gives a self-contained bound

$$F(k, n) \leq (2^{k+1} - 1)(2^k + k! - 1).$$

2. Coefficient compression improves this to $F(k, n) \leq v(k)C_k \leq (k + 1)v(k)^2$, with explicit numerical bounds for $4 \leq k \leq 8$.

3. Classical Waring estimates imply $F(k, n) = O(k^3(\log k)^2)$ uniformly over n.

4. The quadratic case has the exact uniform maximum $\sup_n F(2, n) = 4$.

The cited MathOverflow post contains the earlier nine-term quadratic construction and a specialized proof of $F(3, n) \leq 37$. They are acknowledged as prior material and are not reproduced here. The bounds for $k \geq 4$ in the table are not asserted to be optimal. Improving the coefficient costs C_k and the easier-Waring bounds $v(k)$ remains a natural direction for further work.

A Reproducing the coefficient-cost table

The following short Python program recomputes the elementary coefficient bounds used in the numerical table. It verifies the arithmetic of the displayed C_k bounds, but it does not prove the published bounds for $v(k)$, which are sourced in the references.

```
from math import comb

# Upper bounds for v(k) used in the paper.
v_upper = {2: 3, 3: 5, 4: 10, 5: 10,
           6: 14, 7: 62, 8: 28}

for k, v in v_upper.items():
```

```
    coeffs = [comb(k + 1, j) for j in range(1, k + 2)]
    costs = [min(c, 2**k - c + 1, v) for c in coeffs]
    C = sum(costs)
    print(k, coeffs, costs, C, v * C)
```

B References

References

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